Lowest Common Manager

AlgoExpert · Hard · Recursion

The problem

- In an org chart tree, the lowest manager having both reports somewhere beneath (or being one of them).
- Post-order counting of found reports.

Intuition

- Post-order counts how many of the two reports each subtree contains; the first manager with both is the answer.

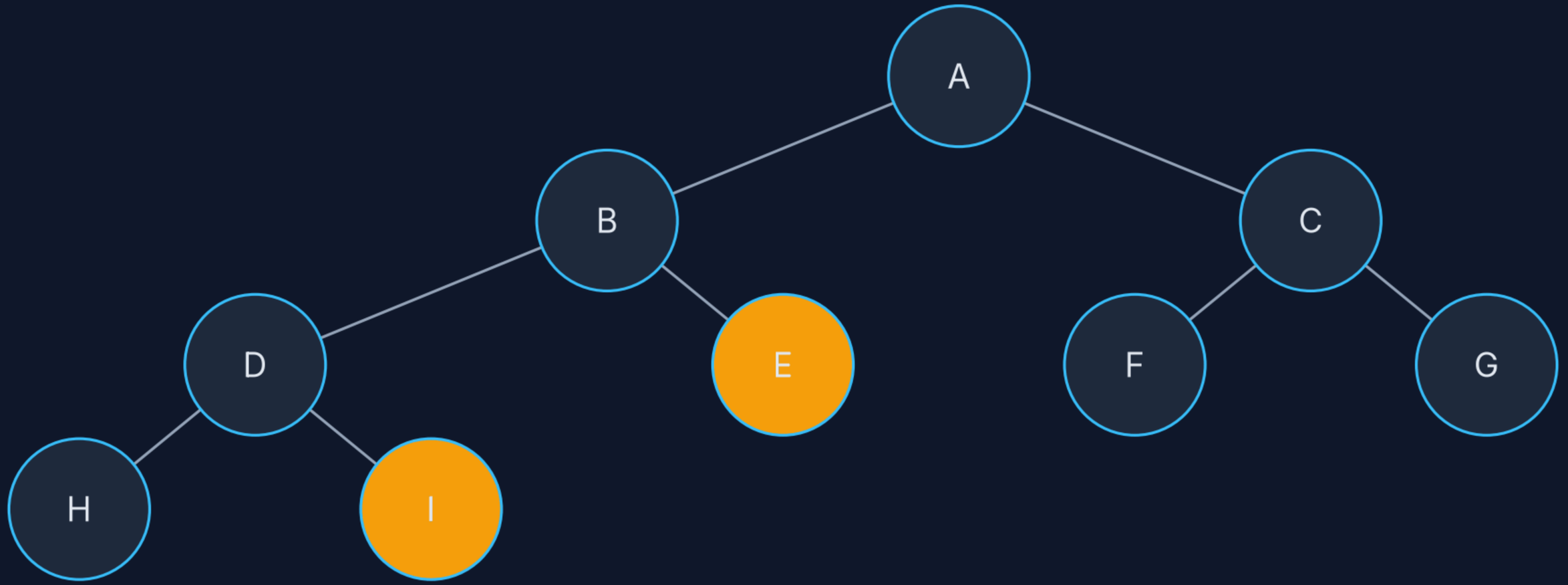
Brute force vs optimal

- Find each report's path from top, compare → O(n) + paths
- Single post-order counting reports → O(n)
- Return the first node reaching count 2

Approach

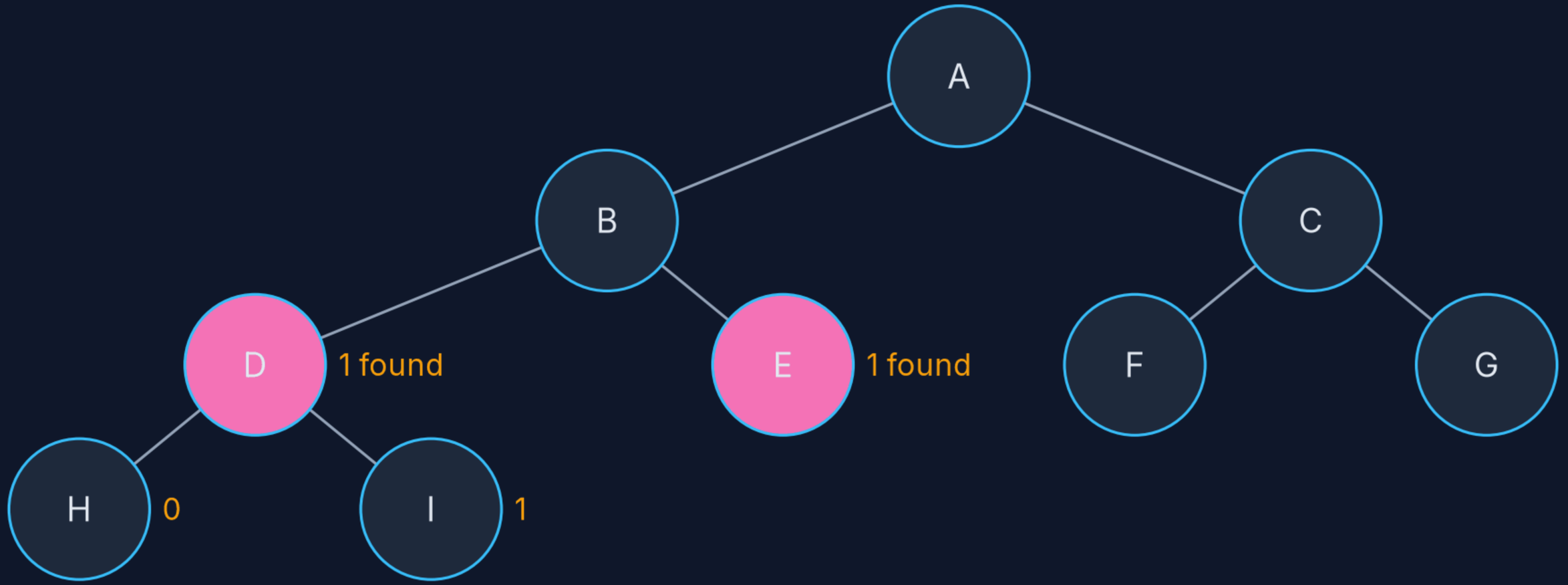
- 1. Recurse into reports
- 2. Count matches in each subtree
- 3. The first node reaching two is the answer
- 4. Return it upward

Worked example



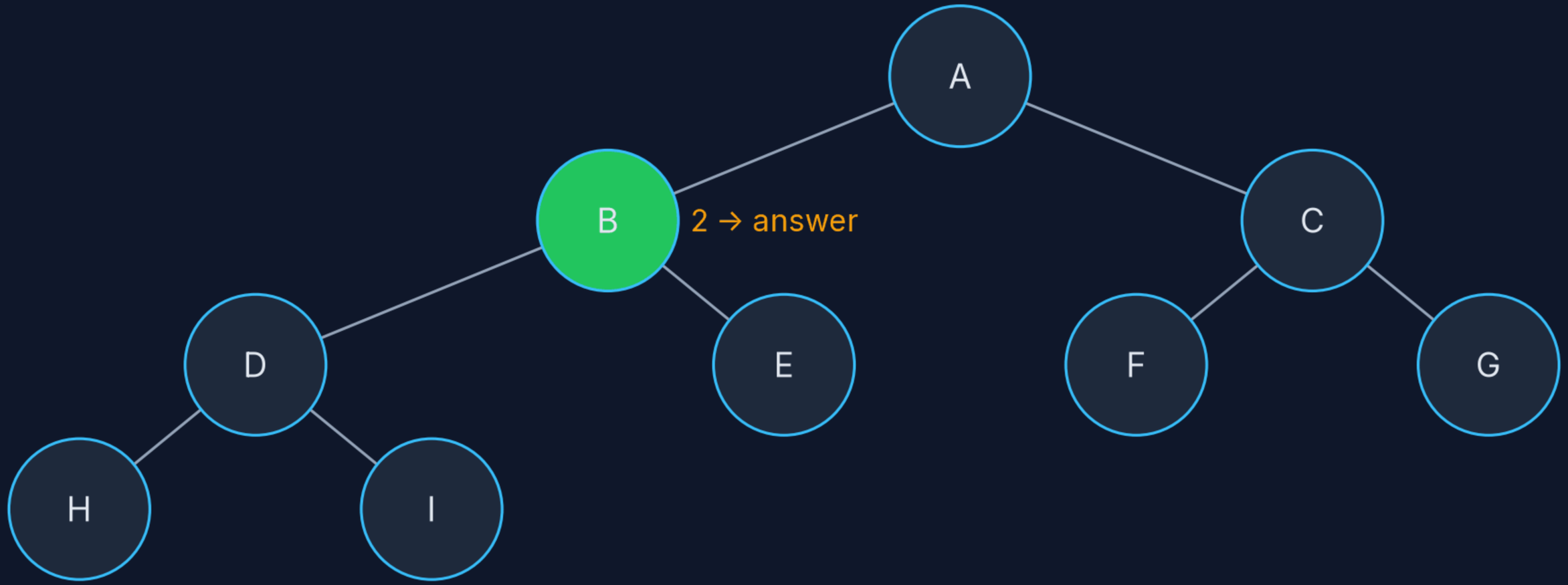
Reports E and I: who's their lowest common manager?

Worked example



Post-order: each manager returns how many reports it covers

Worked example



First manager with count 2 → B

Java solution

```java
public class LowestCommonManager {
    public static class OrgChart {
        public final char name;
        public final List<OrgChart> directReports = new ArrayList<>();
        public OrgChart(char name) { this.name = name; }
        public OrgChart add(OrgChart... r) { directReports.addAll(List.of(r)); return this; }
    }

    public static OrgChart getLowestCommonManager(OrgChart top, OrgChart a, OrgChart b) {
        OrgChart[] ans = {null};
        count(top, a, b, ans);
        return ans[0];
    }

    private static int count(OrgChart m, OrgChart a, OrgChart b, OrgChart[] ans) {
        int n = (m == a ? 1 : 0) + (m == b ? 1 : 0);
        for (OrgChart r : m.directReports) { n += count(r, a, b, ans); if (ans[0] != null) return 2; }
        if (n == 2 && ans[0] == null) ans[0] = m;
        return n;
    }
}
```

Complexity

- Time: O(n)
- Space: O(d)

Edge cases & pitfalls

- A report may be the manager of the other
- Org charts are n-ary
- Stop after found

Interview tips & follow-ups

- Relate to LCA in binary trees

Recap

- Post-order counts how many of the two reports each subtree contains; the first manager with both is the answer.
- Complexity: O(n)